

The Segment LID

Computing the Longest Certified Safe Step Towards the PPO Proposal

Abstract

PSPO currently certifies an axis-aligned parameter orthotope — a Local Invariant Domain (LID) — around the last certified policy, and projects the PPO proposal into it. This note specifies the alternative in which the LID is collapsed onto the *line segment* joining the last certified parameter vector to the proposal, represented exactly as a rank-one zonotope. We state the resulting optimisation problem, prove that it is sound, monotone and well-posed, and show that it is solved to ε -optimality by bisection on a single scalar — replacing a $2n$ -variable primal-dual program that carries no optimality guarantee.

1 From an orthotope to a segment

Let $\theta_k \in \mathbb{R}^n$ be the last certified actor parameters, let $\bar{\theta}_{k+1}$ be the PPO proposal, and let

$$\Delta_k = \bar{\theta}_{k+1} - \theta_k \tag{1}$$

be the proposed update. The current LID is an orthotope $\Theta_k = [\ell_k, u_k] \ni \theta_k$ grown by the directional primal-dual program of the PSPO appendix (§“Directional safe-LID construction”), and the next iterate is $\theta_{k+1} = \text{Proj}_{\Theta_k}^{\ell_2}(\bar{\theta}_{k+1})$.

The geometric difficulty is that an orthotope is a product set. The smallest box containing both θ_k and $\bar{\theta}_{k+1}$ is the axis-aligned hull

$$\text{hull}(\theta_k, \bar{\theta}_{k+1}) = \{\theta_k + \text{diag}(t)\Delta_k : t \in [0, 1]^n\}, \tag{2}$$

so *any* LID that accepts the proposal unchanged must also certify every one of the uncountably many mixed parameter vectors in Eq. (2), of which the straight path

$$[\theta_k, \bar{\theta}_{k+1}] = \{\theta_k + t\Delta_k : t \in [0, 1]\}, \quad t \in [0, 1], \tag{3}$$

is a one-dimensional, measure-zero diagonal. The optimiser proposes a single direction, and the points of Eq. (3) are the candidate iterates along it; certifying the whole hull in order to reach them is pure over-approximation. Two caveats are worth stating at the outset. The orthotope is not *confined* to the segment — its projection may leave the line and exploit n directions of freedom, which partly compensates for the looser bound (see §10). And the tightness gain below concerns the *certificate*, not automatically the step: a direction along which the policy genuinely becomes unsafe quickly admits no long certified step under any abstraction.

Table 1 quantifies the cost on PSPO’s own bound propagation for a three-layer tanh actor: the mean width of the certified logit interval when the same segment is bounded (i) as a rank-one zonotope, (ii) as a rank-one zonotope split into eight sub-segments, and (iii) by interval propagation over the hull of Eq. (2). The reference column is the true range of the logits along the segment, sampled at 2001 points.

Two facts drive everything that follows. First, the segment abstraction is 40–50× tighter than the hull. Second — and more importantly — the segment is one-dimensional, so splitting it into sub-segments drives the relaxation error to zero, which no amount of splitting can achieve for an n -dimensional box.

Table 1: Mean certified logit-interval width for the same segment under three abstractions. Smaller is tighter; the true range is the unattainable optimum.

$\ \Delta_k\ $	true range	rank-one zonotope	eight sub-segments	interval hull
0.05	0.0022	0.0022	0.0022	0.0913
0.25	0.0129	0.0140	0.0129	0.4691
1.00	0.0394	0.0594	0.0397	2.0297

2 The segment as a rank-one zonotope

A zonotope in parameter space is the image of a coefficient box under an affine map,

$$\mathcal{Z} = \left\{ c + \sum_{i=1}^r z_i g_i : z_i \in [\underline{z}_i, \bar{z}_i] \right\}, \quad (4)$$

with centre $c \in \mathbb{R}^n$ and generators $g_i \in \mathbb{R}^n$. Unlike an orthotope, the coefficients z_i are *shared* across every coordinate of θ , and therefore across every layer of the network; this is what lets correlated parameter motion cancel during bound propagation instead of compounding.

Proposition 1 (Exactness of the rank-one representation). *For $0 \leq a \leq b$ define*

$$c(a, b) = \theta_k + \frac{a+b}{2} \Delta_k, \quad g(a, b) = \frac{b-a}{2} \Delta_k, \quad (5)$$

and let $\mathcal{Z}(a, b) = \{c(a, b) + zg(a, b) : z \in [-1, 1]\}$. Then

$$\mathcal{Z}(a, b) = \{\theta_k + t\Delta_k : t \in [a, b]\}. \quad (6)$$

Proof. Substituting Eq. (5), $c(a, b) + zg(a, b) = \theta_k + \left(\frac{a+b}{2} + z\frac{b-a}{2}\right)\Delta_k$. The affine map $z \mapsto \frac{a+b}{2} + z\frac{b-a}{2}$ is a bijection from $[-1, 1]$ onto $[a, b]$ when $a < b$, and maps $[-1, 1]$ onto $\{a\}$ when $a = b$. $\square$

Proposition 1 is the crux: the abstraction introduces *no* geometric error. Every point the certificate must cover is in $\mathcal{Z}(a, b)$, and every point of $\mathcal{Z}(a, b)$ is a point the certificate must cover. All remaining conservatism comes from bound propagation through the network, not from the shape of the set.

Remark 2 (Why the symmetric parameterisation). The segment could equally be written with centre θ_k , generator Δ_k and coefficient $z \in [0, 1]$. The symmetric form of Eq. (5) is preferred because the implementation’s zonotope product absorbs second-order terms $z^2 g \otimes g$ into the centre as an interval, evaluated with interval arithmetic on $z \cdot z$. Over $[0, 1]$ that yields a second-order contribution of magnitude $\|\Delta_k\|^2 \cdot \text{width}([0, 1]) = \|\Delta_k\|^2$, whereas over $[-1, 1]$ with the halved generator it yields $\frac{1}{4}\|\Delta_k\|^2 \cdot \text{width}([-1, 1]) = \frac{1}{2}\|\Delta_k\|^2$. The symmetric form is tighter by a factor of two at no cost.

3 The certification operator

Let $\mathcal{S}_{\text{cert}} \subseteq \mathcal{S}$ be the finite set of certified states, $x(s)$ the policy input of state s , $\mathcal{A}_{\text{safe}}(s)$ the shield-admissible actions, and $z_{\theta,a}(s)$ the actor logits, exactly as in the PSPO appendix. Bound propagation over a parameter set Θ returns sound logit bounds

$$z_{\theta,a}(s) \in [\underline{z}_a(s; \Theta), \bar{z}_a(s; \Theta)] \quad \text{for every } \theta \in \Theta, \text{ every } s, \text{ every } a, \quad (7)$$

where for a zonotope $\Theta = \mathcal{Z}$ the propagation carries the shared coefficient z through every layer and concretises only at the output.

The returned certificate is the *hard* per-state test. In the default all-safe semantics,

$$c_s(\Theta) = \mathbf{1} \left[\min_{a \in \mathcal{A}_{\text{safe}}(s)} z_a(s; \Theta) > \max_{b \notin \mathcal{A}_{\text{safe}}(s)} \bar{z}_b(s; \Theta) \right], \quad (8)$$

which requires every admissible action to dominate every inadmissible one; in the weaker any-safe semantics the first min is replaced by a max, which only requires the greedy action to be admissible. Both are supported; the analysis below is identical for either, and we write c_s for whichever is in force.

3.1 The sub-segment and partitioned certificates

For a sub-segment $[a, b] \subseteq [0, 1]$ define the scalar certificate

$$C(a, b) = \min_{s \in \mathcal{S}_{\text{cert}}} c_s(\mathcal{Z}(a, b)) \in \{0, 1\}, \quad (9)$$

i.e. $C(a, b) = 1$ exactly when a single bound-propagation pass certifies every policy on the sub-segment on every certified state. For a step length $\alpha \in [0, 1]$ and a partition $0 = t_0 < t_1 < \dots < t_K = \alpha$ into K equal parts, define the *partitioned certificate*

$$C_K(\alpha) = \prod_{i=1}^K C(t_{i-1}, t_i) \in \{0, 1\}, \quad t_i = \frac{i}{K} \alpha. \quad (10)$$

The partition is a proof device only: the certified object is the union $\bigcup_i \mathcal{Z}(t_{i-1}, t_i) = \mathcal{Z}(0, \alpha)$, a single segment.

4 The optimisation problem

4.1 The ideal problem

Write $\mu_\theta(s) = \arg \max_a z_{\theta, a}(s)$ for the greedy policy. What we would like to compute is the longest prefix of the proposed step on which the greedy policy is shield-admissible everywhere:

$$\alpha^* = \max \left\{ \alpha \in [0, 1] : \mu_{\theta_k + t \Delta_k}(s) \in \mathcal{A}_{\text{safe}}(s) \quad \forall s \in \mathcal{S}_{\text{cert}}, \forall t \in [0, \alpha] \right\}. \quad (11)$$

Problem (11) is not directly solvable. The constraint quantifies over a continuum of t ; the predicate is an arg max comparison, hence piecewise constant, discontinuous and non-convex in t ; and evaluating it at finitely many t certifies nothing about the intervals in between.

4.2 Problem P: the certified relaxation

Replacing the exact predicate by the sound certificate of Eq. (10) gives the problem PSPO actually solves:

$$\boxed{\hat{\alpha} = \max \left\{ \alpha \in [0, 1] : C_K(\alpha) = 1 \right\}, \quad \theta_{k+1} = \theta_k + \hat{\alpha} \Delta_k.} \quad (12)$$

The decision variable is the single scalar α . The certified LID returned to the rest of the pipeline is the rank-one zonotope $\mathcal{Z}(0, \hat{\alpha})$, with centre $\theta_k + \frac{\hat{\alpha}}{2} \Delta_k$, generator $\frac{\hat{\alpha}}{2} \Delta_k$ and coefficient interval $[-1, 1]$.

5 Properties

Proposition 3 (Soundness). *If $C_K(\alpha) = 1$ then every θ on the segment $[\theta_k, \theta_k + \alpha\Delta_k]$ satisfies $\mu_\theta(s) \in \mathcal{A}_{\text{safe}}(s)$ for all $s \in \mathcal{S}_{\text{cert}}$. Consequently $\hat{\alpha} \leq \alpha^*$, and the enforced iterate θ_{k+1} of Eq. (12) is certified safe.*

Proof. Let $\theta = \theta_k + t\Delta_k$ with $t \in [0, \alpha]$. The partition covers $[0, \alpha]$, so $t \in [t_{i-1}, t_i]$ for some i , and by Proposition 1, $\theta \in \mathcal{Z}(t_{i-1}, t_i)$. Since $C_K(\alpha) = 1$ we have $C(t_{i-1}, t_i) = 1$, so $c_s(\mathcal{Z}(t_{i-1}, t_i)) = 1$ for every s . By the soundness of bound propagation, Eq. (7), the logits of θ lie inside the propagated bounds, and Eq. (8) then forces every admissible logit of θ above every inadmissible one; in particular the greedy action is admissible. The inequality $\hat{\alpha} \leq \alpha^*$ follows because every α feasible for Eq. (12) is feasible for Eq. (11). $\square$

Proposition 4 (Monotonicity). *C_K is non-increasing in α : if $0 \leq \alpha' \leq \alpha \leq 1$ and $C_K(\alpha) = 1$, then $C_K(\alpha') = 1$. Hence the feasible set of Eq. (12) is the interval $[0, \hat{\alpha}]$.*

Proof. Each sub-segment of the α' partition is contained in $\mathcal{Z}(0, \alpha') \subseteq \mathcal{Z}(0, \alpha)$, and is therefore contained in the union of the α sub-segments. It suffices to show that the certificate is inclusion-monotone, i.e. that $\mathcal{Z}_1 \subseteq \mathcal{Z}_2$ implies $[\underline{z}_a(s; \mathcal{Z}_1), \bar{z}_a(s; \mathcal{Z}_1)] \subseteq [\underline{z}_a(s; \mathcal{Z}_2), \bar{z}_a(s; \mathcal{Z}_2)]$. This holds layer by layer: an affine layer maps the shared coefficient interval through a fixed linear form, so shrinking the interval shrinks each generator contribution; and the ReLU and tanh relaxations are constructed from the concretised input range, which is nested by induction, so their slopes and residual intervals are those of a nested domain. Nested logit bounds make Eq. (8) monotone in the same direction. $\square$

Proposition 5 (Well-posedness). *$C_K(0) = 1$ whenever θ_k is certified safe. In particular Eq. (12) is always feasible and $\hat{\alpha}$ is well defined.*

Proof. At $\alpha = 0$ every sub-segment degenerates to $\mathcal{Z}(0, 0) = \{\theta_k\}$, whose generator is zero. Bound propagation over a degenerate zonotope is exact: the concretised range at each layer has zero width, so the ReLU and tanh relaxations are evaluated at a point and their residual intervals collapse. Hence $\underline{z}_a(s; \mathcal{Z}(0, 0)) = \bar{z}_a(s; \mathcal{Z}(0, 0)) = z_{\theta_k, a}(s)$, and Eq. (8) reduces to the exact greedy safety test at θ_k , which holds by the induction hypothesis of the PSPO safety invariant. $\square$

Corollary 6 (Bisection solves Problem P). *By Propositions 4 and 5, bisection on $[0, 1]$ started from the bracket $(\underline{\alpha}, \bar{\alpha}) = (0, 1)$ returns α with $\hat{\alpha} - \varepsilon \leq \alpha \leq \hat{\alpha}$ after $\lceil \log_2(1/\varepsilon) \rceil$ evaluations of C_K , each costing K bound-propagation passes over $\mathcal{D}_{\text{sh}}$.*

Proposition 7 (Refinement). *For K' a multiple of K , $C_K(\alpha) = 1 \Rightarrow C_{K'}(\alpha) = 1$; hence $\hat{\alpha}$ is non-decreasing in the number of splits, and remains bounded above by α^* .*

Proof. Each sub-segment of the K partition is a union of sub-segments of the K' partition, and the certificate is inclusion-monotone by the argument of Proposition 4. The upper bound is Proposition 3. $\square$

Proposition 7 is what makes the segment LID qualitatively different from the orthotope LID. Splitting is a branch-and-bound refinement in *one* dimension: the relaxation error of each nonlinearity shrinks with the width of the sub-segment, so K is a dial that trades bound-propagation passes for reach, and Table 1 shows $K = 8$ already recovering the exact range. The analogous refinement for an n -dimensional box would require splitting along n coordinates and is combinatorially hopeless.

6 Relation to the projection-distance objective

Problem (12) is stated as a feasibility problem in α , but it is equivalent to the projection objective PSPO already implements. Since $\theta_{k+1} = \theta_k + \alpha \Delta_k$,

$$\|\bar{\theta}_{k+1} - \theta_{k+1}\|_2 = (1 - \alpha) \|\Delta_k\|_2, \quad (13)$$

which is strictly decreasing in α . Maximising the certified step length is therefore identical to minimising the distance between the enforced iterate and the PPO proposal subject to certified safety,

$$\min_{\theta \in \mathcal{Z}(0, \alpha)} \|\bar{\theta}_{k+1} - \theta\|_2^2 \quad \text{s.t.} \quad C_K(\alpha) = 1, \quad (14)$$

i.e. Problem P is the exact one-dimensional specialisation of the `projection_distance` LID objective. Note also that Euclidean projection onto a rank-one zonotope is closed-form: for any θ ,

$$\text{Proj}_{\mathcal{Z}(a,b)}^{\ell_2}(\theta) = c + z^* g, \quad z^* = \text{clamp}\left(\frac{\langle \theta - c, g \rangle}{\|g\|_2^2}, -1, 1\right), \quad (15)$$

the one-dimensional analogue of the coordinate-wise clipping used for orthotopes. No iterative projection solver is required.

7 Algorithm

Algorithm 1 states the resulting procedure. The first test of each pass is the fast path: if the full proposal certifies, PSPO accepts it unchanged and the enforcement costs a single certificate evaluation. The **while** loop is the bisection of Corollary 6. The outer **loop** applies Proposition 7: if bisection returns nothing, the partition is refined and both the fast path and the bisection are retried on the tighter proof, up to $K_{\max}$. The final branch fails closed, exactly as the orthotope path does when no checkpoint is certified.

Two cost details are worth stating. The fast path means that an update which is already safe is never slowed down by the machinery: one pass and the proposal is taken. And because bisection tests midpoints, a step certified only after refinement is returned to within ε rather than exactly, so ε should be set at the scale below which the difference in step length is immaterial (the default 10^{-3} costs ten passes).

Algorithm 1 Longest certified safe step towards the PPO proposal

Require: last certified θ_k ; proposal $\bar{\theta}_{k+1}$; data $\mathcal{D}_{\text{sh}}$; tolerance ε ; initial splits K ; split cap $K_{\max}$

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1:  $\Delta_k \leftarrow \bar{\theta}_{k+1} - \theta_k$ 
2: loop
3:   if  $C_K(1) = 1$  then
4:     return  $\hat{\alpha} = 1, \mathcal{Z}(0, 1)$  ▷ proposal accepted unchanged
5:    $(\underline{\alpha}, \bar{\alpha}) \leftarrow (0, 1)$ 
6:   while  $\bar{\alpha} - \underline{\alpha} > \varepsilon$  do
7:      $\alpha \leftarrow \frac{1}{2}(\underline{\alpha} + \bar{\alpha})$ 
8:     if  $C_K(\alpha) = 1$  then
9:        $\underline{\alpha} \leftarrow \alpha$ 
10:    else
11:       $\bar{\alpha} \leftarrow \alpha$ 
12:    if  $\underline{\alpha} > 0$  or  $K \geq K_{\max}$  then
13:      break
14:     $K \leftarrow 2K$  ▷ refine the proof, Proposition 7
15: if  $\underline{\alpha} = 0$  then
16:   return FAIL ▷ revert to  $\theta_k$ 
17: return  $\hat{\alpha} = \underline{\alpha}, \mathcal{Z}(0, \hat{\alpha})$  with centre  $\theta_k + \frac{\hat{\alpha}}{2} \Delta_k$ , generator  $\frac{\hat{\alpha}}{2} \Delta_k$ , coefficients  $[-1, 1]$ 
```

8 What this replaces

Table 2 contrasts the two LID computations. The essential change is that an n -dimensional constrained program with no optimality certificate is replaced by a one-dimensional monotone feasibility problem solved to a stated tolerance.

Table 2: The orthotope LID program against Problem P.

	Orthotope LID	Segment LID (Problem P)
Decision variables	$2n$ bounds (ℓ_k, u_k)	one scalar α
Solver	augmented-Lagrangian primal–dual, Adam on the primal, fixed iteration budget	bisection
Optimality	none; the returned box is the last certified checkpoint visited	ε -optimal for the given K
Cost	200 bound-propagation passes with gradients	$\lceil \log_2(1/\varepsilon) \rceil$ gradient-free passes, each $\times K$
Surrogate	smooth LSE margin plus temperature calibration	none; the hard certificate is used directly
Set certified	axis-aligned hull, Eq. (2)	the segment itself, Eq. (3)
Refinable	no	yes, by Proposition 7
Projection	coordinate-wise clipping	closed form, Eq. (15)

Because the search never differentiates the certificate, the temperature calibration of the appendix — and the failure mode in which calibration finds no admissible β_k and the update is rejected outright — disappears from this path entirely. The smooth surrogate exists only to give the primal–dual optimiser a gradient; bisection needs no gradient.

9 Relation to the withdrawn line search

An earlier PSPO fallback performed a bisection line search on the same segment, accepting the largest probed α whose *interpolated policy* passed a pointwise greedy check. That procedure is unsound: passing the check at finitely many probes says nothing about the intervals between them, and the strategy was withdrawn as a confirmed defect. Problem P has the same search structure and the same cost profile, but each probe evaluates $C_K(\alpha)$ — a certificate over the whole sub-segment — rather than a point test. It therefore recovers the reach of line search while retaining the guarantee of projection: by Proposition 3, the accepted point is not merely safe, the entire path to it is.

10 Measured behaviour

Two things were measured on FrozenLake with a single-safe-action shield, a linear actor and PPO at three learning rates, comparing the segment LID against the directional orthotope LID at its default 200-iteration budget.

The certificate is near-tight. For each enforcement the true maximal safe step α^* of Eq. (11) was located by dense bisection on the exact greedy predicate and compared against the certified $\hat{\alpha}$:

K	1	2	4	8	16
mean $\hat{\alpha}/\alpha^*$	0.909	0.947	0.971	0.984	0.985
worst $\hat{\alpha}/\alpha^*$	0.637	0.784	0.881	0.930	0.930
cost (ms)	12	19	35	78	137

So Problem P is solved to within a few percent of the unattainable optimum, and refinement behaves exactly as Proposition 7 predicts: monotone, with cost linear in K and returns that flatten past $K = 8$. $K = 4$ is the implementation default.

The cost advantage is large; the reach advantage is not. Each LID took 17–35 ms against 620–730 ms for the orthotope, a 20–40 $\times$ saving, and every run ended greedy-safe. But on the objective of Eq. (13) — the residual distance $\|\bar{\theta}_{k+1} - \theta_{k+1}\|$ — the two are close to a wash, the segment better at some learning rates and worse at others, though its best-case enforcements are consistently closer to the proposal.

This is not a defect in the certificate, which the table above shows is nearly exact. It is the geometric cost of the restriction: the orthotope may project off the line and use n coordinates of freedom to land nearer the proposal, whereas the segment must buy its entire displacement along one direction. The honest summary is that the segment LID buys a near-exact, refinable, gradient-free certificate at a fraction of the cost, not automatically a longer step — and that the thickened segment of §11 is the natural way to recover the missing degrees of freedom.

11 Limitations

The LID has no volume. $\mathcal{Z}(0, \hat{\alpha})$ is one-dimensional, so it cannot absorb a subsequent update in any other direction. The segment must be recomputed at every enforcement, and a mode that computes one LID and reuses it across many updates is not meaningful for this geometry. The compensating fact is that recomputation is roughly two orders of magnitude cheaper than the primal–dual growth it replaces.

The Rashomon reading weakens. An orthotope LID can be described as a set of near-optimal safe models, which supports statements about the neighbourhood of the certified policy. A segment supports no such statement: the guarantee it delivers is per-iterate — every enforced θ_{k+1} , and the whole path taken to reach it, is certified — which is precisely what Theorem 1 of the appendix requires, and nothing more.

A middle ground exists. Nothing forces rank one. Taking $g_1 = \frac{\hat{\alpha}}{2}\Delta_k$ together with $r - 1$ further generators yields a thickened segment that retains directional reach while recovering some volume for reuse, at the cost of reintroducing a search over the generators. Problem P is the $r = 1$ corner of that family, and the corner at which the problem becomes a monotone scalar feasibility test.